Better evidence on secondary prophylaxis could change Uganda's scale-up decision, but rarely Indonesia's

Pillar 3 · What would better evidence be worth? Value of information for RHD prophylaxis, 2026–2050

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We value perfect knowledge of four prophylaxis inputs for one decision

One decision, 2026–2050

Current care, or scale up echo screening plus secondary antibiotic prophylaxis (SAP)?

Four uncertain SAP inputs

1. Efficacy, stage A → B (minimal → mild)
2. Efficacy, stage B → C (mild → advanced)
3. Age to which protection lasts
4. Protection with partial adherence

Held fixed

Disease model, care cascade, costs, valuation. No SAP effect on late stages or on deaths.

One trial of 818 participants anchors the effect of prophylaxis on early RHD

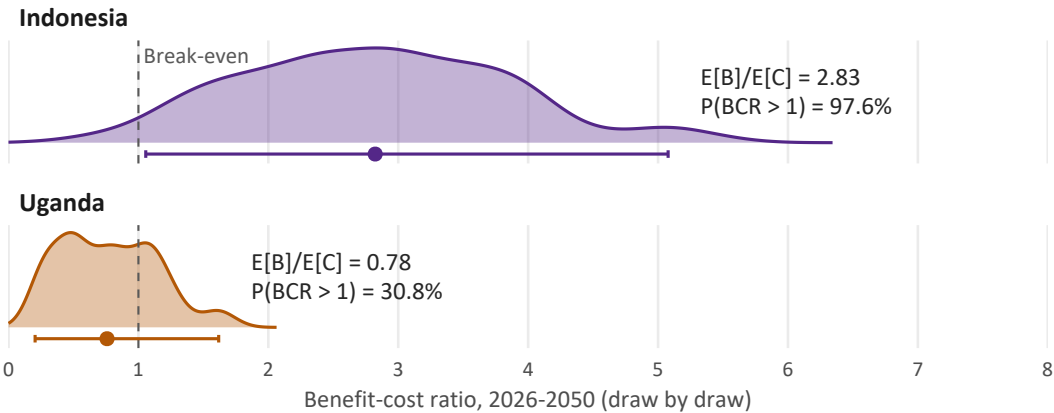
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risk ratio for progression of early RHD on prophylaxis (95% CI 0.03–0.29)

Effects at other stages, older ages and partial adherence are extrapolations

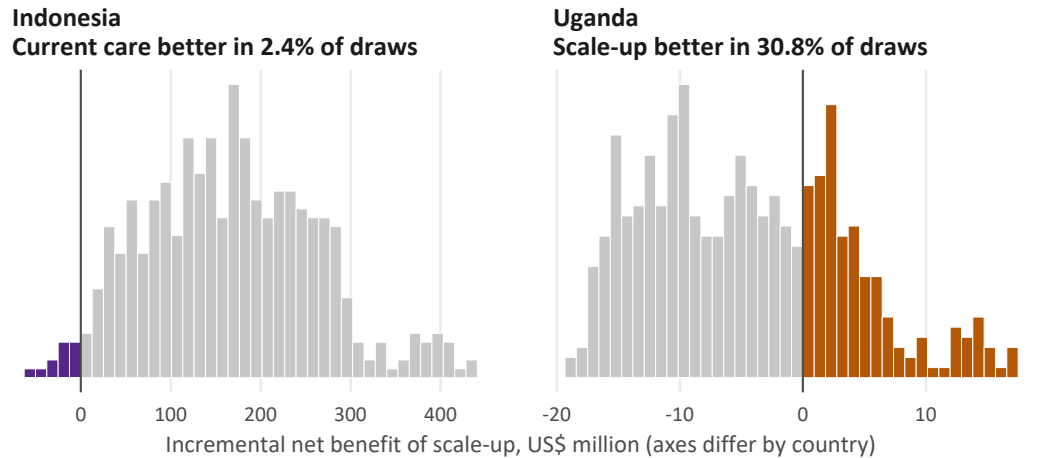
Systematic review of long-term antibiotic prophylaxis (Bray et al., 2024); the trial is GOAL (Beaton et al., 2022).

By 2050, scale-up returns US$2.8 per dollar in Indonesia, US$0.78 in Uganda



Incremental benefit (deaths averted × reference VSL) ÷ incremental programme cost, discounted 3% to 2026, constant 2025 US$, 2026–2050 only. Bars: median and 95% PSA interval, 500 draws.

Perfect information pays only in draws where the other option wins



Colour: draws in which the option not chosen on current evidence is better. EVPI = average net benefit regained in those draws. VSL × 1, 2026–2050, discounted, constant 2025 US$.

In Indonesia, perfect evidence is worth at most US$0.50 million

The value of knowing how well secondary prophylaxis works · Indonesia

US$0.50m

expected value of perfect information, 2026–2050

Scale-up preferred on current evidence (BCR 2.83); better in 97.6% of draws

Perfect information compared with	US$ (2025)	Share
Programme cost, 2026–2050	US$678m	**0.073%**
Govt. health spending (GGHE-D), 1 year (2023)	US$19.1bn	**0.0026%**
GDP, 1 year (2023)	US$1,320bn	**0.000038%**

EVPI bootstrap 95%: US$0.21m–US$0.86m. Gross programme cost, discounted; GGHE-D and GDP in 2023 (World Bank, 2026; World Health Organization, 2025). Scale comparisons, not budgets.

In Uganda, perfect evidence, worth up to US$1.5 million, could flip the choice

The value of knowing how well secondary prophylaxis works · Uganda

US$1.5m

expected value of perfect information, 2026–2050

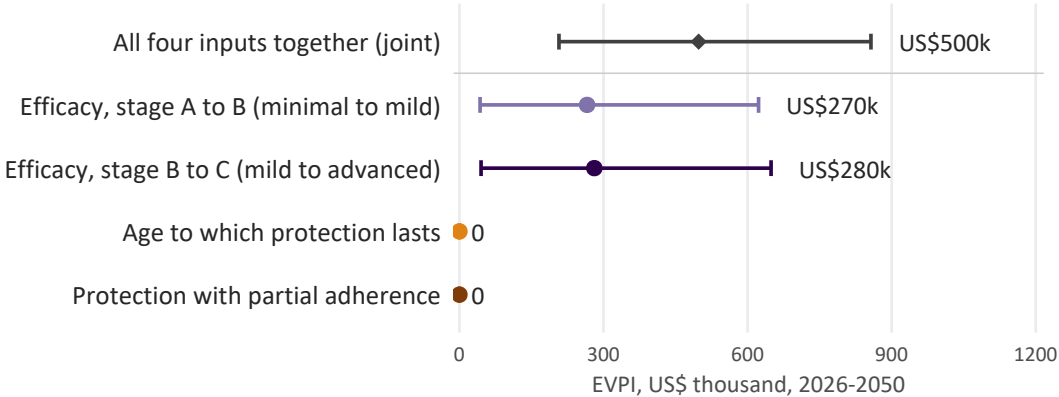
Current care preferred (BCR 0.78); scale-up better in 30.8% of draws

Perfect information compared with	US$ (2025)	Share
Programme cost, 2026–2050	US$255m	**0.60%**
Govt. health spending (GGHE-D), 1 year (2023)	US$502m	**0.31%**
GDP, 1 year (2023)	US$54.4bn	**0.0028%**

EVPI bootstrap 95%: US$1.3m–US$1.8m. Gross programme cost, discounted; GGHE-D and GDP in 2023 (World Bank, 2026; World Health Organization, 2025). Scale comparisons, not budgets.

In Indonesia, only the two efficacy inputs carry information value

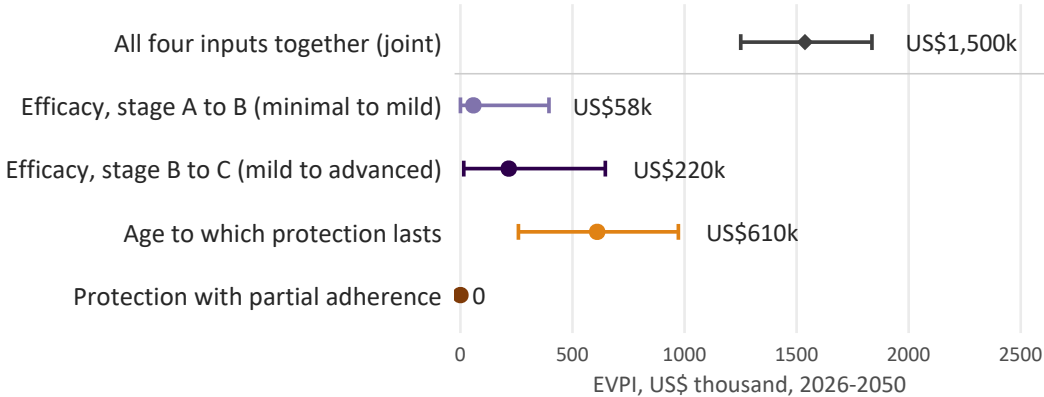
Which uncertainty is worth resolving? · Indonesia



Learning one input alone, others still uncertain; values overlap and do not add. Bars: bootstrap 95%; imprecise at 500 draws.

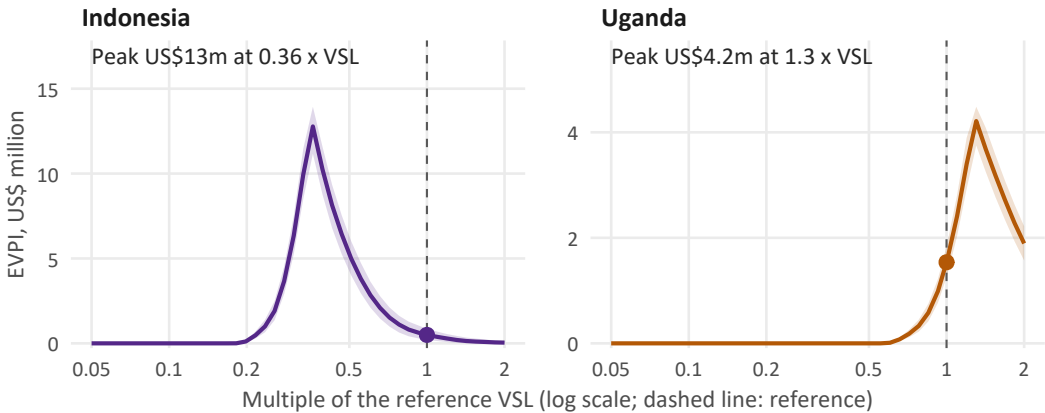
In Uganda, the age to which protection lasts is the most valuable single input

Which uncertainty is worth resolving? · Uganda



The joint value exceeds every single input: combinations of inputs, not one alone, flip the decision. Bars: bootstrap 95%.

Information value peaks where the decision sits near break-even



Joint EVPI with deaths averted valued at a multiple of the reference VSL; y-axes differ by country. Dot: reference. Band: bootstrap 95%.

These are upper bounds from a fixed model with provisional priors

A skeptic would say

Priors are not expert-agreed, the model misses GBD burden, and no study gives perfect information.

What the analysis supports

A cap on what resolving these four inputs is worth for this decision; a real study is worth less.

Not assessed

SAP effects on stage C $\longrightarrow$ D; recalibration; any specific study design.

For this decision, research on these inputs costing more than the cap cannot pay off

	Choice on current evidence	Cap (EVPI)	Most valuable single input
Indonesia	Scale-up (97.6% of draws)	**US$0.50m**	Efficacy, A $\rightarrow$ B or B $\rightarrow$ C (overlap)
Uganda	Current care (69.2% of draws)	**US$1.5m**	Age to which protection lasts

EVPI caps the expected value of any study on these four inputs for the national scale-up decision, 2026–2050; other decisions or countries could add value.

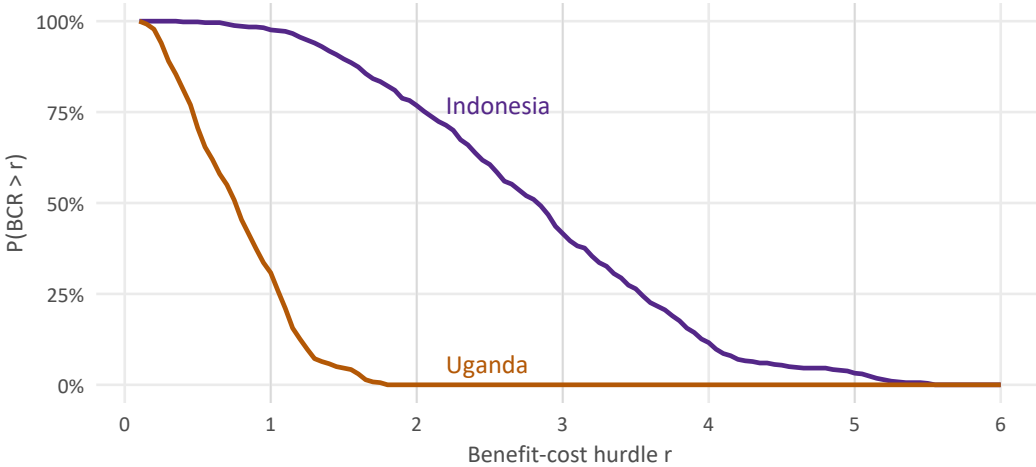
Through 2050, knowing how well prophylaxis works could change Uganda's scale-up decision but rarely Indonesia's, and is worth at most US$1.5 million in Uganda and US$0.50 million in Indonesia.

The BCR is incremental and discounted, and counts only benefits to 2050

	Indonesia	Uganda
Expected benefit E[B]	US$259m	US$16.7m
Expected incremental cost E[C]	US$91.5m	US$21.4m
E[B] / E[C] (decision ratio)	**2.83**	**0.78**
Median BCR (95% PSA interval)	2.82 (1.06–5.08)	0.76 (0.20–1.62)
Mean of draw ratios E[B/C] (descriptive)	2.82	0.77
P(BCR $>$ 1) / P(BCR $>$ 2)	97.6% / 76.8%	30.8% / 0.0%
P(BCR $>$ 3) / P(BCR $>$ 5)	41.6% / 3.2%	0.0% / 0.0%
Point-value BCR (production inputs)	2.38	0.71
Draws with incremental cost $\leq$ 0	0	0

B: deaths averted $\times$ reference VSL; C: scale-up minus reference cost; discounted 3% to 2026, constant 2025 US$, 2026–2050. With C $>$ 0, BCR $>$ 1 $\Leftrightarrow$ INB $>$ 0: the decision the VOI addresses.

Indonesia clears a benefit-cost hurdle of 2 in most draws; Uganda never does



Share of 500 draws with B − r C > 0 (equal to P(BCR > r) since every draw has C > 0); 2026–2050, discounted.
Descriptive: the decision rule is BCR > 1.

A higher benefit-cost hurdle is a different objective, not a larger welfare gain

$$G_r = B - r\,C = r\left[\tfrac{1}{r}B - C\right]$$

Hurdle	Indonesia, US$ million	Uganda, US$ million
r = 2	11 (8.4–13)	0
r = 3	31 (26–35)	0
r = 5	0.69 (0.32–1.1)	0

VOI of the objective G_r (joint, bootstrap 95%). G_r re-weights costs by r, so its VOI is not a welfare value; it equals r times the EVPI at a VSL multiple of 1/r.

Priors on the four inputs are provisional and not expert-agreed

Input	Point	Prior (95% range)	Sampled (median, 95%; or n)
Efficacy, A $\rightarrow$ B	0.45	0.03–0.29 (trial) or $1 - s(1 - RR)$	0.33 (0.04–0.73)
Efficacy, B $\rightarrow$ C	0.45	0.068–0.59 (trial) or = A $\rightarrow$ B	0.38 (0.07–0.75)
Age to which protection lasts	39	20 / 39 / 100 with P 0.25 / 0.5 / 0.25	137 / 222 / 141
Protection with partial adherence	0.50	Beta(4.54, 4.54)	0.52 (0.21–0.78)

Two equally weighted regimes share one indicator: trial efficacy (Beaton et al., 2022; Bray et al., 2024) or benefit only via fewer rheumatic fever recurrences, RR 0.22–0.69 (Bray et al., 2024), scaled by an unelicited share s. Point M from Manyemba and Mayosi (2002); partial adherence after de Dassel et al. (2018).

Information values come from one set of 500 paired draws per country

$$\text{EVPI} = \mathrm{E}_{\theta}\left[\max_{d} \text{NB}(d, \theta)\right] - \max_{d} \mathrm{E}_{\theta}\left[\text{NB}(d, \theta)\right]$$

- Joint value of the four inputs = EVPI, only because nothing else varies here
- Single input: GAM regression of B and C on that input (Strong et al., 2014)
- Single-input values overlap; never summed or stacked (Rothery et al., 2020)
- Intervals: 1,000 bootstrap resamples of whole paired draws (estimator precision)

Calibration, starting state, cascade, costs and valuation are held fixed

Held fixed

- Calibrated natural history and incidence (fit to GBD: inadequate)
- 2026 engine state, care cascade, unit costs
- Reference-case VSL (Robinson et al., 2019)

Not assessed

- SAP effect on C $\rightarrow$ D and on stage-D death (fixed at none)
- Effect waning; recalibration under other SAP values
- Value of a specific study (EVSI)

Efficacy (M) is full-adherence efficacy; achieved adherence and coverage are separate, fixed cascade inputs.
Calibration fit: inadequate (outside its GBD diagnostic tolerances) in both countries, so all results are preliminary.

Scale comparisons use one year of GDP and of domestic government health spending

Constant 2025 US$	Indonesia	Uganda
Gross programme cost, 2026–2050 (discounted)	US$678m	US$255m
GGHE-D, 2023 (one year)	US$19.1bn	US$502m
GDP, 2023 (one year)	US$1,320bn	US$54.4bn

GDP: current local currency (World Bank, 2026). GGHE-D: WHO share of GDP (World Health Organization, 2025) times that GDP; excludes donor and private funds; not a ministry budget. 2023 is the latest common year. Converted with local CPI to 2025 and the 2025 market exchange rate (Turner et al., 2019); not PPP.

In Indonesia, each value is under one-tenth of 1% of programme cost

Input	EVPI, US$ thousand (95%)	P(top)	Programme	GGHE-D, 1 yr	GDP, 1 yr
All four together (joint)	500 (210–860)		0.073%	0.0026%	0.000038%
Efficacy, A → B	270 (43–620)	37.4%	0.039%	0.0014%	0.000020%
Efficacy, B → C	280 (45–650)	62.4%	0.041%	0.0015%	0.000021%
Age to which protection lasts	0 (0–0)	0.0%	0%	0%	0%
Protection with partial adherence	0 (0–0)	0.0%	0%	0%	0%

VSL × 1, 2026–2050, constant 2025 US$. P(top): share of bootstrap replicates in which the input alone ranks first. Shares use one fixed denominator for estimate and bounds. Joint EVPI / expected incremental cost (US$91.5m): 0.54%.

In Uganda, perfect information is worth 0.60% of programme cost

Input	EVPI, US$ thousand (95%)	P(top)	Programme	GGHE-D, 1 yr	GDP, 1 yr
All four together (joint)	1,500 (1,300–1,800)		0.60%	0.31%	0.0028%
Efficacy, A → B	58 (0–400)	0.0%	0.023%	0.012%	0.00011%
Efficacy, B → C	220 (15–650)	2.9%	0.085%	0.043%	0.00040%
Age to which protection lasts	610 (260–970)	97.1%	0.24%	0.12%	0.0011%
Protection with partial adherence	0 (0–0)	0.0%	0%	0%	0%

VSL × 1, 2026–2050, constant 2025 US$. P(top): share of bootstrap replicates in which the input alone ranks first. Joint EVPI / expected incremental cost (US$21.4m): 7.2%.

At low valuations every draw favours current care, so EVPI is zero

Valuation	Indonesia	Uganda
VSL × 0.05	0.0% / 0	0.0% / 0
VSL × 0.1	0.0% / 0	0.0% / 0
VSL × 1 (reference, for contrast)	97.6% / US$0.50m	30.8% / US$1.5m
0.5 × GNI pc per DALY (proxy)	0.0% / 0	0.0% / 0
1 × GNI pc per DALY (proxy)	0.0% / 0	0.0% / 0

Cells: P(scale-up better) / joint EVPI. DALY proxy: discounted deaths averted × 30; GNI per capita 2026 from the production income path. Zero VOI means one option wins in every sampled state, not that nothing is uncertain.

All 500 paired draws were used; single-input values remain imprecise

	Indonesia	Uganda
Draws requested / used / excluded	500 / 500 / 0	500 / 500 / 0
Joint EVPI, bootstrap relative SE	0.35	0.10
Joint EVPI, split halves (US$ million)	0.50 / 0.50	1.6 / 1.5
Single inputs > 0 flagged imprecise	2 of 2	3 of 3
Smallest GAM k-index p-value	0.23	0.30
Bootstrap resamples (failed fits)	1,000 (0)	1,000 (0)

VSL × 1, 2026–2050. Imprecise: bootstrap relative SE above 0.20. Extending beyond 500 draws (same run, same prior) would tighten them; not done here.

The joint value moves with the weight given to trial-level efficacy

Prior weight on trial efficacy	Indonesia	Uganda
w = 0.25	US$0.74m	US$0.89m
w = 0.5	US$0.50m	US$1.5m
w = 0.75	US$0.25m	US$2.2m

Joint EVPI at VSL × 1, 2026–2050, re-weighting the same draws (importance weights; base case w = 0.5). Exploratory.

Scale-up averts about 550 deaths in Indonesia and 250 in Uganda by 2050

Mean (95% PSA interval)	Indonesia	Uganda
RHD deaths averted (undiscounted)	550 (200–1,030)	250 (60–540)
DALYs averted, proxy (discounted)	9,600 (3,600–17,800)	4,200 (1,100–9,100)
Gross scale-up cost, US$ million (discounted)	678 (672–692)	255 (254–257)

2026–2050, national population, 500 draws. DALY proxy: discounted deaths averted × 30 (no age-specific years of life lost, no disability); not modelled DALYs.

References

Beaton, A., Okello, E., Rwebembera, J., Grobler, A., Engelman, D., Alepere, J., Canales, L., Carapetis, J., DeWyer, A., Lwabi, P., Mirabel, M., Mocumbi, A. O., Murali, M., Nakitto, M., Ndagire, E., Nunes, M. C., Omara, I. O., Sarnacki, R., Scheel, A., … Steer, A. C. (2022). Secondary Antibiotic Prophylaxis for Latent Rheumatic Heart Disease. *New England Journal of Medicine*, *386*(3), 230–240. https://doi.org/10.1056/NEJMOA2102074

Bray, J. J. H., Thompson, S., Seitler, S., Ali, S. A., Yiu, J., Salehi, M., Ahmad, M., Pelone, F., Gashau, H., Shokraneh, F., Ahmed, N., Cassandra, M., Marijon, E., Celermajer, D. S., & Providencia, R. (2024). Long-term antibiotic prophylaxis for prevention of rheumatic fever recurrence and progression to rheumatic heart disease. *Cochrane Database of Systematic Reviews*, *2024*(9), Article CD015779. https://doi.org/10.1002/14651858.CD015779

de Dassel, J. L., de Klerk, N., Carapetis, J. R., & Ralph, A. P. (2018). How many doses make a difference? An analysis of secondary prevention of rheumatic fever and rheumatic heart disease. *Journal of the American Heart Association*, *7*(24), Article e010223. https://doi.org/10.1161/JAHA.118.010223

Manyemba, J., & Mayosi, B. M. (2002). Penicillin for secondary prevention of rheumatic fever [[Heart]]. *Cochrane Database of Systematic Reviews*, (3). https://doi.org/10.1002/14651858.CD002227

References

Robinson, L. A., Hammitt, J. K., & O'Keeffe, L. (2019). Valuing mortality risk reductions in global benefit-cost analysis. *Journal of Benefit-Cost Analysis*, *10*(S1), 15–50. https://doi.org/10.1017/bca.2018.26

Rothery, C., Strong, M., Koffijberg, H. (, Basu, A., Ghabri, S., Knies, S., Murray, J. F., Schmidler, G. D. S., Steuten, L., & Fenwick, E. (2020). Value of Information Analytical Methods: Report 2 of the ISPOR Value of Information Analysis Emerging Good Practices Task Force. *Value in Health*, *23*(3), 277–286. https://doi.org/10.1016/j.jval.2020.01.004

Strong, M., Oakley, J. E., & Brennan, A. (2014). Estimating multiparameter partial expected value of perfect information from a probabilistic sensitivity analysis sample: A nonparametric regression approach. *Medical Decision Making*, *34*(3), 311–326. https://doi.org/10.1177/0272989X13505910

Turner, H. C., Lauer, J. A., Tran, B. X., Teerawattananon, Y., & Jit, M. (2019). Adjusting for inflation and currency changes within health economic studies. *Value in Health*, *22*(9), 1026–1032. https://doi.org/10.1016/j.jval.2019.03.021

World Bank. (2026). *World Development Indicators*. Retrieved November 10, 2026, from https://databank.worldbank.org/source/world-development-indicators

References

World Health Organization. (2025). *Global Health Expenditure Database*. Retrieved November 10, 2026, from https://apps.who.int/nha/database